# Tosin Kolajo GBADEGESIN

Applied Microeconomics, Predictive ML, Causal ML, Forecasting, and Game Theory
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## EDUCATION

**Ph.D. Economics**, Howard University, Washington DC, USA – *March 2026*
**M.Sc. Economic Development**, University of Ibadan, Ibadan, Nigeria – *Sept 2018*
**B.Sc. Microbiology**, Obafemi Awolowo University, Ile-Ife, Nigeria – Sept *2013*

## PROFESSIONAL TRAINING & CERTIFICATES

**Artificial Intelligence with Python**, Harvard University Extension School – *2023*
**Post Graduate Program in Data Science**, Career Era & Birchwood University, Orlando, Florida – *2022*
**Data Analytics**, Google Career Certificate Program – *2021*
**Machine Learning Engineering**, Amazon Web Services Certificate Program – *2021*
**Blum Global Fellowship in Development Finance**, Hebrew University Business School & Jerusalem Institute for Policy Research, Israel – *2020*

## TECHNICAL SKILLS

**Econometrics & Causal Inference**
Panel & Time-Series Models; IV & GMM; Spatial Econometrics; DiD; RDD; PSM; Fixed Effects; Synthetic Control; Quantile Regression; Causal ML (CATE, Causal Forest, DML, DTE, CDTE)

**Data Science & Machine Learning**
Exploratory Data Analysis; Predictive Modeling; Ensemble methods (Boosting: XGBoost, LightGBM; Bagging: Random Forest; Stacking: Super Learner), Model Evaluation & Optimization; NLP & LLMs; Deep Learning; Experimental Design, A/B Testing; Feature Engineering.

**Programming & Tools**
Python, R; Stata; SQL; Power BI; Tableau; MATLAB; ArcGIS; Google Earth; HTML/JavaScript; GitHub; Azure; AWS.

## HONORS & AWARDS

**Frederick Douglass Doctoral Scholar**, Howard University Graduate School, 2023–2024
**World Bank Group Africa Doctoral Fellow –** selected as 1 of 32 fellows from over 3,000 applicants, 2023
**Blum Global Fellow in Development Finance** – Jerusalem Institute for Policy Research, Israel, 2020
**Early Career Conference Grant**, Association of Commonwealth Universities (ACU), London, UK – 2019
**Outstanding Research Contribution Award**, *Young African Leaders Journal of Development*, Ethiopia – 2018
**Young WASH Scholar**, Africa Water Association (AfWA) & USAID West Africa – 2018
**YALI Leadership Award** in Energy and Environmental Sustainability, Nigeria – 2018
**Young IFAD Scholar,** International Fund for Agricultural Development (IFAD) – 2018

## JOB MARKET PAPER

The Building Energy Performance Standard Program: Impacts on Building's Energy Efficiency and GHG Emissions in Washington, DC [Link]

## PUBLICATIONS

- "Climate Change and Energy Dynamics: Evidence from BRICS and G7 Countries," *Energy & Environment*, 2024, 0(0). [Link]
- "Evaluating the Economic Impact of the G20 Compact Initiative: Evidence from Causal Inference Using Advanced Machine" (with N. Yameogo), *Journal of Sustainable Development*, 2024, 17(5), 56–76. [Link]
- "Defense Expenditure and National Well-Being: Evidence from the G7 Countries," *Defence and Peace Economics*, 2024, 1–12. [Link]
- "Effectiveness of Collective Action in Reducing Transaction Cost for Smallholder Paddy Farmers in Tanzania" (with L. Popoola), *African Journal of Economic and Sustainable Development*, 2020, 7(4), 391–404. [Link]
- "Water Availability Challenges in Low-Income Areas of Agbowo Community, Ibadan, Nigeria" (with O. Olayide), *International Journal of Circular Economy and Waste Management*, 2021, 1(1), 81–96. [Link]
- "Assessment of Women and Children Vulnerability to Water Use in Low-Income Urban Area of Agbowo Community, Ibadan, Nigeria" (with O. Olayide), *International Journal of Sustainability Management and Information Technologies*, 2019, 5(2), 29–34. [Link]
- "Transaction Cost Factors Affecting the Economic Life of Smallholder Paddy Farmers in Tanzania" (with L. Popoola and L. Muhoni), *Africa Journal of Sustainable Development*, 2019, 8(1). [Link]

## World Bank Group Papers

- "Climate Shocks and Their Effects on Food Security, Prices, and Agricultural Wages in Afghanistan" (with B. P. J. Andree and A. Braimoh), *Policy Research Working Paper, World Bank Group (PLANET)*, 2024. [Link]
- "Reproducibility Package for Climate Shocks and Their Effects on Food Security, Prices, and Agricultural Wages in Afghanistan" (with B. P. J. Andree), *World Bank Policy Research Working Papers*, 2024. [Link]

## World Bank Group Reports

- International Finance Corporation. (2025). Closing the Gap: A Data Outlook on Women Economic Opportunities. [Link]

- International Finance Corporation. (2024). *Repurposing power markets: The path to sustainable and affordable energy for all*. International Finance Corporation. [Link]
- The World Bank Group. (2023). *Compact monitoring report: Africa Advisory Group meeting, November 2023*. G20 Compact with Africa. [Link]

**Book Chapters**

- "A Review on Impact of Changing Climate on Sustainable Food Consumption," in *Global Food Politics and Approaches to Sustainable Consumption*, 2020, 54–76. [Link]
- "Climate Change Implication on Airborne Infections: Roadmap for Nigeria's Health Sector Development," in *Handbook of Climate Change Resilience*, 2018, 1–10. [Link]

**Blog**

- "Better Data for Better Outcomes: New Tools and Analysis to Break Afghanistan's Food Security Crisis" (with B. P. J. Andree and A. Braimoh), *World Bank Blogs*, 2025. [Link]

**Working Paper**

- Heterogeneous Effects of Cooling Demand on Household Electricity Consumption using Causal ML Models, [Link]
- Impact of Climate-Driven Cooling Demand on U.S. Household Electricity Consumption and Expenditure
- Beyond Affordability: A Spatial Econometric Analysis of Renter Cost Burdens and Social Determinants in the United States

**CONFERENCES & SEMINARS**

**Annual NABE Tech Economics Conference,** NABE, Seattle WA, USA – 2025
**Annual Economic Measurement Seminar**, NABE, Washington DC, USA – 2025
**Empirical Methods in Energy Economics**, EMEE Network & The World Bank, Washington DC, USA – 2025
**American Economic Association / National Economic Association Conference (ASSA Meetings)** – 2024
**Development Studies Association Conference**, Trinity College, Dublin, Ireland – 2019

**EMPLOYMENT**

**World Bank Group** – Economist
Washington, DC | 01/2023 – Present

- Lead development of World Bank's Private Sector Gender & Economic Inclusion Data Hub, designing indicators, statistical frameworks, automated data pipelines and BI dashboards processing 2–4 TB of multi-source data, while coordinating harmonization, validation, and stakeholder engagement to support policy analysis and global reporting.
- Developed model-based early warning systems for food insecurity using econometric and VAR approaches, improving lead times for Early Response Financing by up to 12 months.
- Led Compact with Africa's economic impact report demonstrating 5–10% gains in FDI and domestic financing, and delivering analyses that informed the Germany-Africa Head of State Summit and World Bank Annual Meetings.
- Applied econometrics and machine learning methods (DiD, TMLE, PSM, Random Forest, LightGBM, Gradient Boosting) to evaluate policy and investment impacts, integrating statistical analysis and causal inference design.
- Conducted applied research on energy and extractives markets, refining IFC's AIMM framework to assess development impact and linking power market structure to sector outcomes.
- Built ML models (using HistGradient Boosting, LightGBM, CatBoost, XGBoost) on 500+ GB IFC's historical debt data, improving IFC's non-performing loan detection by 15% and strengthening debt portfolio monitoring.
- Engaged senior stakeholders across World Bank Group, academia, private sector, and government, translating complex data-driven findings into compelling narratives and strategic policy recommendations.

**Howard University Graduate School** – Research Assistant | Teaching Assistant
Washington, DC | 2024 – 2025

- Processed over 100 million trade records from UN Comtrade for faculty research on international trade.
- Teaching Assistant for Economic Statistics (Undergraduate): grading, tutorials, and data-lab instruction.

**Edelman Global Advisory** – Research Analyst (Washington, DC | 09/2022 – 12/2022)

- Conducted economic and regulatory research on trade and climate policy across six regions, generating insight reports that guided policy actions impacting markets representing over $500B in annual trade.

**DC Sustainable Energy Utility (DCSEU) & DC Department of Energy and Environment (DOEE)**
Energy Efficiency Analyst | Washington, DC | 06/2022 – 12/2022

- Analyzed 100+ GB of building benchmarking data and performed benefit-cost and impact analyses showing 5–10% performance gains.
- Developed guidance materials and statistical reports for decarbonization pilots, affordable housing retrofits, and the Solar for All program.
- Delivered data-driven insights and interactive dashboards by designing automated analytics pipelines and responding to stakeholder inquiries on renewable energy programs.

**Nigerian Economic Summit Group** – Bridge Policy Research Fellow (Nigeria | 08/2020 – 08/2021)

- Analyzed COVID-19 household impact data and reviewed national development plans in water resources and sanitation to inform policy reforms.
- Co-led the establishment of the UN CEO Water Mandate in Nigeria and contributed to medium-term national development frameworks and Agenda 2050.

- Provided technical input at the Nigerian Economic Summit, advancing gender mainstreaming and digital innovation strategies for private sector growth.

**Jerusalem Institute for Policy Research | Gigawatt Global Inc.** – Development Research Specialist
Israel | 01/2020 – 09/2020

- Assessed pre-feasibility and feasibility studies for grid-connected solar projects, analyzing solar yield, transmission infrastructure, financial structuring, and ESG impacts.
- Designed and optimized solar transmission and interconnection lines using Google Earth and PVSYST, conducting irradiation and energy demand analyses to inform project planning.
- Built financial models and mobilized project financing, engaging investors, development finance institutions, and diplomatic stakeholders to secure capital and trade support.

**International Institute for Local Development** – Data Analyst (Abuja, Nigeria | 09/2019 – 07/2021)

- Applied statistical analysis and machine learning (Random Forest, AdaBoost, KNN, SVM, CNN) to evaluate World Bank–funded rural access and agricultural projects, generating insights for decision-making.
- Processed, cleaned, and visualized data using Excel, R, and Tableau, ensuring accuracy and integrity across large datasets.
- Produced reports and presentations that translated analytical findings into actionable recommendations for project stakeholders.

**Clean Edge Exclusive Solution** – Team Lead, Projects & Programs (Nigeria | 06/2017 – 12/2020)

- Directed teams deploying mini-grid solar and energy efficiency projects, while coordinating sustainability initiatives with over 60 volunteers.
- Designed and implemented evaluation frameworks to measure project performance, alignment with objectives, and community impact.
- Co-convened Nigeria's first Annual Youth Summit on Environment and led a national assessment of plastic circularity in partnership with the World Bank Group and PwC.

**Africa Water Association & USAID West Africa** – Research Fellow (Nigeria | 07/2018 – 12/2018)

- Researched urban water supply challenges in Nigeria, designing and managing data collection pipelines with Kobo Collect survey tool, performing statistical analyses in SPSS and R, and generating actionable insights.
- Developed interactive Tableau dashboards and automated reporting workflows using SQL and Python, presenting results at an international sustainable development conference.

**International Fund for Agricultural Development** – Research Fellow (Arusha, Tanzania | 01/2018 – 06/2018)

- Analyzed survey data using R to identify factors influencing smallholder farmers' market access, applying econometric techniques to generate actionable insights.
- Authored and published findings in peer-reviewed journals, including the Africa Journal of Sustainable Development, translating technical results into policy-relevant recommendations.

**Crown FM Broadcast Group & Independent Media** – Research Analyst (Nigeria | 06/2013 – 06/2016)

- Led data-driven market research initiatives, integrating SQL-based querying and Python analytics to assess audience behavior and content performance across digital platforms.
- Developed interactive dashboards and performance metrics (engagement) to inform strategic content decisions and enhance multi-platform reach.